30 MCQ Questions: Networking Fundamentals

Comprehensive Single-Page Quick Review Sheet

1. What does the acronym LAN stand for?
A) Local Access Network
B) Local Area Network
C) Large Area Network
D) Linked Access Node
Answer: B) Local Area Network

2. How many layers are in the OSI Model?
A) 4 Layers
B) 5 Layers
C) 7 Layers
D) 6 Layers
Answer: C) 7 Layers

3. Which layer of OSI handles IP routing?
A) Data Link
B) Network Layer
C) Transport Layer
D) Session Layer
Answer: B) Network Layer (Layer 3)

4. What is the main function of ARP?
A) Resolve IP to MAC
B) Resolve Name to IP
C) Encrypt traffic
D) Assign IP address
Answer: A) Resolve IP to MAC Address

5. Which protocol resolves domain names to IP addresses?
A) DHCP
B) DNS
C) HTTP
D) FTP
Answer: B) DNS

6. Which protocol automatically assigns IP addresses?
A) DNS
B) DHCP
C) ICMP
D) SNMP
Answer: B) DHCP

7. What is the length of an IPv4 address?
A) 16 bits
B) 32 bits
C) 64 bits
D) 128 bits
Answer: B) 32 bits

8. What is the length of an IPv6 address?
A) 32 bits
B) 64 bits
C) 128 bits
D) 256 bits
Answer: C) 128 bits

9. A MAC address operates at which OSI layer?
A) Physical
B) Data Link
C) Network
D) Transport
Answer: B) Data Link (Layer 2)

10. What is the standard size of a MAC address?
A) 32 bits
B) 48 bits
C) 64 bits
D) 128 bits
Answer: B) 48 bits (6 Bytes)

11. Which device connects different networks together?
A) Switch
B) Hub
C) Router
D) Repeater
Answer: C) Router

12. A network device that broadcasts to all ports is a:
A) Switch
B) Hub
C) Router
D) Gateway
Answer: B) Hub

13. Which port number is used by HTTP?
A) Port 21
B) Port 53
C) Port 80
D) Port 443
Answer: C) Port 80

14. Which port number is used by HTTPS?
A) Port 80
B) Port 22
C) Port 443
D) Port 25
Answer: C) Port 443

15. TCP is best described as a protocol that is:
A) Connectionless
B) Connection-oriented
C) Unreliable
D) Physical-layer
Answer: B) Connection-oriented

16. UDP is best described as a protocol that is:
A) Connectionless
B) Connection-oriented
C) High reliability
D) Heavy overhead
Answer: A) Connectionless (Fast & Unreliable)

17. Which utility tests end-to-end network reachability?
A) `nslookup`
B) `ping`
C) `ipconfig`
D) `netstat`
Answer: B) ping (uses ICMP)

18. What protocol does the `ping` command use?
A) TCP
B) UDP
C) ICMP
D) IGMP
Answer: C) ICMP

19. In Star topology, all devices connect directly to a:
A) Central Hub/Switch
B) Single Cable
C) Ring loop
D) Next device
Answer: A) Central Hub or Switch

20. What topology connects every device to every other device?
A) Bus
B) Ring
C) Mesh
D) Star
Answer: C) Full Mesh

21. What is the loopback IPv4 address for local host testing?
A) 192.168.1.1
B) 127.0.0.1
C) 10.0.0.1
D) 255.255.255.255
Answer: B) 127.0.0.1

22. What default subnet mask is for a Class C network?
A) 255.0.0.0
B) 255.255.0.0
C) 255.255.255.0
D) 255.255.255.255
Answer: C) 255.255.255.0 (/24)

23. Which protocol is used for secure remote command line access?
A) Telnet
B) SSH
C) FTP
D) SMTP
Answer: B) SSH (Port 22)

24. What protocol is primarily used to send emails?
A) POP3
B) IMAP
C) SMTP
D) SNMP
Answer: C) SMTP (Port 25)

25. What media type uses light pulses for data transfer?
A) Coaxial
B) Twisted Pair
C) Fiber Optic
D) Wireless radio
Answer: C) Fiber Optic

26. Data unit at the Transport Layer is called a:
A) Bit
B) Frame
C) Packet
D) Segment
Answer: D) Segment (or Datagram for UDP)

27. Data unit at the Network Layer is called a:
A) Packet
B) Frame
C) Segment
D) Bit
Answer: A) Packet

28. Data unit at the Data Link Layer is called a:
A) Bit
B) Frame
C) Packet
D) Message
Answer: B) Frame

29. What technique translates private IPs to a public IP?
A) NAT
B) DNS
C) DHCP
D) VPN
Answer: A) NAT (Network Address Translation)

30. Simultaneous communication in both directions is:
A) Simplex
B) Half Duplex
C) Full Duplex
D) Multiplex
Answer: C) Full Duplex